Lecture 37: Transformer: Electrical Circuit

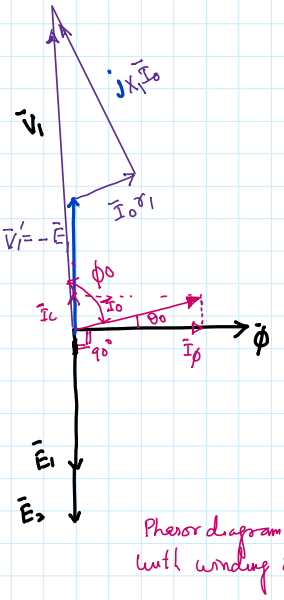
1. Ideal transformer
2. Transformer at no load

The leakage of flux is accounted as voltage drop with the reactive element $X_{l_1} = X_e$. The reactance X_{e1} is denoted as leakage reactance of the primary windi Now with this the phasor diagram is shown below.

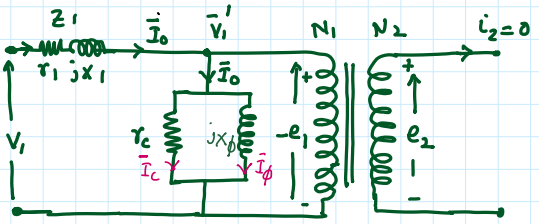
$$\bar{S}_1 = \bar{V}_1 \bar{I}_0^* = V_1 (I_0 \angle \phi_0)^* = V_1 I_0 \cos\phi_0 + jV_1 I_0 \sin\phi_0$$
$$= P_1 + jQ_1$$

$$P_1 = V_1 I_0 \cos\phi_0 = I_1^2 r_1 + P_C$$

$$Q_1 = V_1 I_0 \sin\phi_0 = I_0^2 X_1 + I_\phi^2 X_\phi$$



Phasor diagram at no load with winding resistance

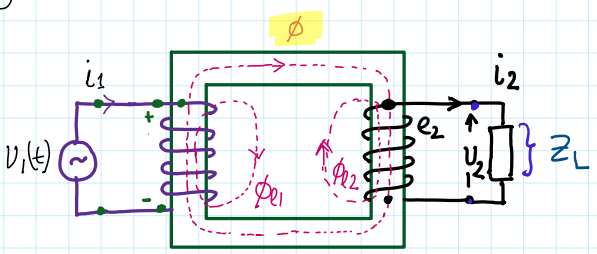


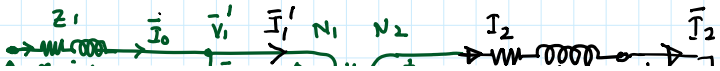
The voltages and currents are related as

$$\begin{cases} \bar{V}_1 = (-\bar{E}_1) + \bar{I}_0 (r_1 + jX_1) \\ \bar{V}_1 = \bar{V}_1' + \bar{I}_0 (r_1 + jX_1) \\ \bar{V}_2 = \bar{E}_2 \end{cases}$$

Transformer at Load: The transformer is supplying load connected at the secondary. The current i_2 flows ($\bar{I}_2$ as phasor equivalent).

This setsup ampere turns in secondary winding, i.e., $N_2 I_2$.





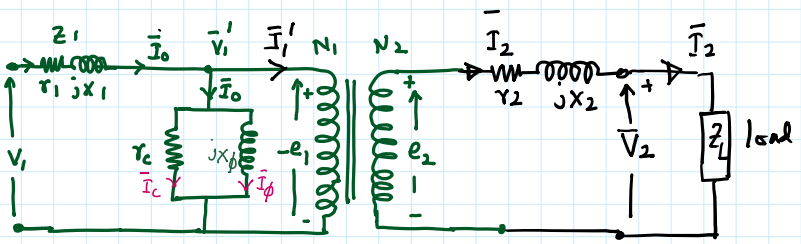


Fig.: Circuit under load condition

The sequence of events are: $\bar{E}_2 \rightarrow \bar{V}_2 \rightarrow \bar{I}_2 \rightarrow N_2\bar{I}_2$ (Ampere Turns)

$$\bar{I}_2 = \frac{\bar{V}_2}{\bar{Z}_L} = \frac{V_2\angle 0}{|Z_2|\angle\theta_2} = \frac{V_2}{|Z_2|}\angle -\theta_2$$

$N_2\bar{I}_2$ = Ampere Turns on secondary Side
= Magnetomotive force
= MMF

EMF = Electromotive force

$$N_1\bar{I}_1' + N_2\bar{I}_2 = 0$$

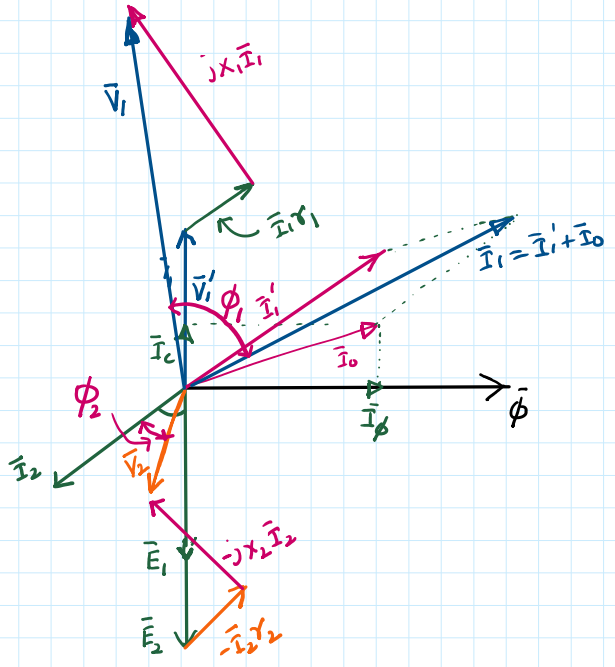
$$\bar{I}_1' = -\frac{N_2}{N_1}\bar{I}_2 = \frac{N_2}{N_1}(-\bar{I}_2)$$

$$\bar{I}_1 = \bar{I}_0 + \bar{I}_1'$$

$$\bar{V}_1 = (-\bar{E}_1) + \bar{I}_1(r_1 + jx_1)$$

$$\bar{V}_2 = \bar{E}_2 - (r_2 + jx_2)\bar{I}_2 \;;$$

$$\bar{I}_2 = \frac{\bar{V}_2}{\bar{Z}_L}$$



Phasor diagram of transformer under load. ($\bar{I}_2 \neq 0$)

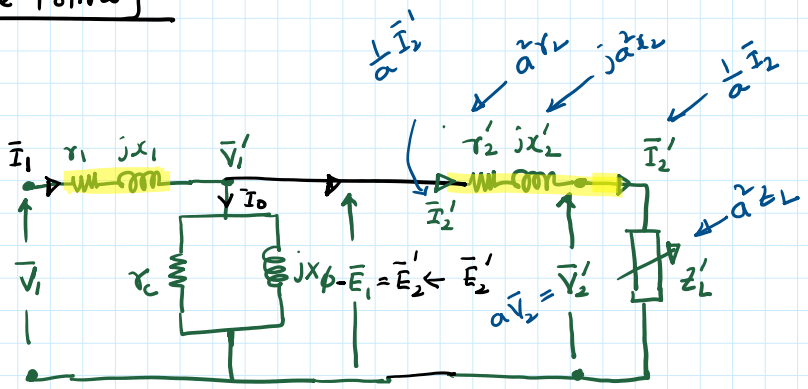
Equivalent Circuit of Transformer:

(A) Secondary Referred to Primary, $\bar{V}_2 \rightarrow \bar{V}_2'$, $\bar{I}_2 \rightarrow \bar{I}_2'$, $\bar{Z}_L \rightarrow \bar{Z}_L'$, $\bar{E}_2 \rightarrow \bar{E}_2'$

(B) Primary Referred to secondary $\bar{E}_1 \rightarrow \bar{E}_1'$, $I_1 \rightarrow \bar{I}_1'$, $z_1 \rightarrow z_1'$, $X_\phi \rightarrow X_\phi'$, $r_c \rightarrow r_c'$

(B) Primary Referred to secondary $\bar{E}_1 \to \bar{E}_1'$, $I_1 \to \bar{I}_1'$, $z_1 \to z_1'$, $X_\phi \to X_\phi'$, $r_c \to r_c'$

Secondary Refered to the Primary



$\bar{E}_2 \to \bar{E}_2' \to (-\bar{E}_1)$

$$E_2' = E_1 = \frac{N_1}{N_2} E_2$$

$$\boxed{\bar{E}_2' = \left(\frac{N_1}{N_2}\right) \bar{E}_2} \quad -(*)$$

$$E_2' I_2' = E_2 I_2$$

$$I_2' = \frac{E_2}{E_2'} * I_2 = \frac{N_2}{N_1} I_2$$

$$\boxed{\bar{I}_2' = \frac{N_2}{N_1} \bar{I}_2}$$

Same losses: $I_2'^2 r_2' = I_2^2 r_2$

$$r_2' = \left(\frac{I_2}{I_2'}\right)^2 r_2 = \left(\frac{N_1}{N_2}\right)^2 r_2$$

$$r_2' = \left(\frac{N_1}{N_2}\right)^2 r_2 = a^2 r_2$$

↑ Transformer Ratio = Turn Ratio

$$I_2'^2 x_2' = I_2^2 x_2 \Rightarrow \boxed{x_2' = \left(\frac{N_1}{N_2}\right)^2 x_2 = a^2 x_2}$$

$$z_2' = r_2' + jx_2' = a^2 z_2 = \left(\frac{N_1}{N_2}\right)^2 z_2$$

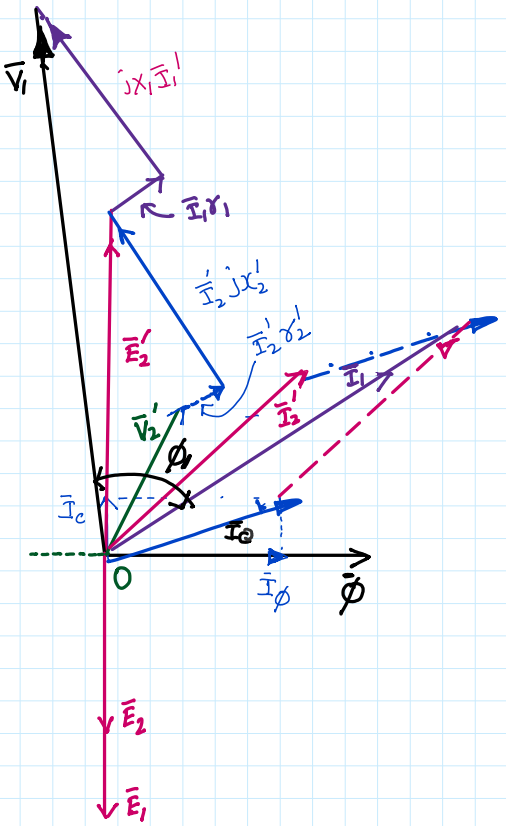
$$\bar{V}_2' = \left(\frac{N_1}{N_2}\right) \bar{V}_2$$

$$z_L' = \left(\frac{N_1}{N_2}\right)^2 z_L = a^2 z_L$$

$I_0 << I_{rated}$ (1500 A)

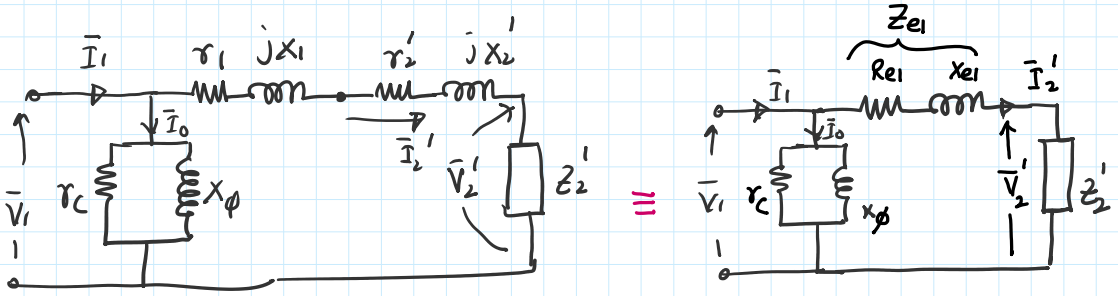
$\simeq$ 5-6% of the I_{rated}

Phasor Diagram of equivalent circuit referred to primary

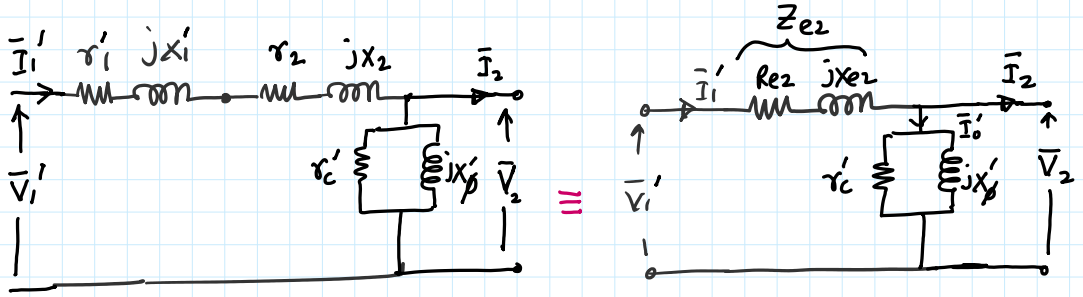


$(+X_2')$

Approximate Equivalent Circuit

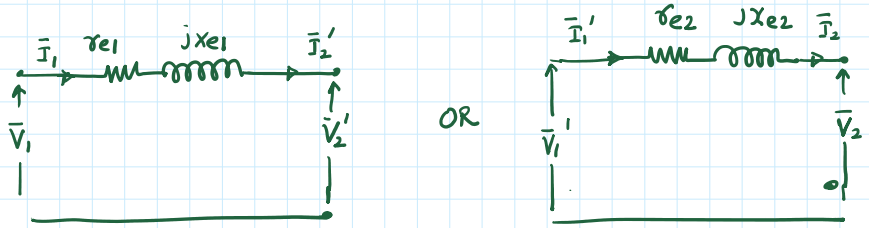


Approximate Equivalent ckt of TIF referred to primary



Approximate Equivalent ckt of TIF referred to secondary

To simplify even more, the shunt branch is neglected. The simplified approximate equivalent circuit for the transformer is:



Referred to primary

Referred to secondary

Here: $r_{e1} = r_1 + r_2' = r_1 + a^2 r_2$

$x_{e1} = x_1 + x_2' = x_1 + a^2 x_2$

$a = N_1/N_2$

$r_{e2} = r_2 + r_1' = r_2 + \left(\frac{1}{a}\right)^2 r_1$

$x_{e2} = x_2 + x_1' = x_2 + \left(\frac{1}{a}\right)^2 x_1$

Primary winding impedance: $Z_1 = r_1 + jX_1$

Secondary winding impedance: $Z_2 = r_2 + jX_2$

Primary winding impedance referred to secondary, $Z_1' = r_1' + jX_1'$

$Z_1' = \left(\frac{N_2}{N_1}\right)^2 Z_1$

Secondary winding impedance referred to primary $Z_2' = r_2' + jX_2'$

$Z_2' = \left(\frac{N_1}{N_2}\right)^2 Z_2$

$Z_{e1} = r_{e1} + jX_{e1} = (r_1 + r_2') + j(X_1 + X_2')$

= Total impedance of transformer's

$Ze_1 = \dots$

= Total impedance of transformer's approximate equivalent ckt referred to primary

$Ze_2 = r_{e2} + jX_{e2} = (r_1' + r_2) + j(x_1' + x_2)$

= Total impedance of transformer's approximate equivalent ckt referred to secondary.